

Compare Ratios

Lesson 3-5

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

\$3 per 1 pound → unit rate is \$3/lb

Unit rate

A rate for just 1 of something, like cost for 1 item.

4:6 and 2:3 are equivalent (both simplify to 2:3)

Equivalent ratio

Two ratios that mean the same thing.

3:4 vs. 2:5 — which has more of the first ingredient per unit?

Compare

To look at ratios and see which is bigger, smaller, or equal.

8:12 → divide both by 4 → 2:3

Simplify

To make a ratio as small as possible while keeping the same comparison.

To compare $\frac{3}{5}$ and $\frac{4}{7}$, use LCD 35: $\frac{21}{35}$ vs $\frac{20}{35}$

Common denominator

A bottom number that two fractions can share so you can compare them.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can compare ratios by finding unit rates or using equivalent ratios.

1 Notice

Chef Academy is holding a Flavor Challenge!

2 Key idea

I can compare ratios by finding unit rates or using equivalent ratios.

3 Words I'll use

Unit rate

Equivalent ratio

Compare

Simplify

Common denominator



Think About It

- What quantities are being compared in each recipe?
- Can you tell just by looking which recipe has more cocoa per ounce?
- What would make the comparison fair?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: Chef A uses 2 cups of cheese for every 8 crackers. Chef B uses 3 cups of cheese for every 9 crackers. Who uses more cheese per cracker?

- A. Chef B
- B. Chef A
- C. They use the same amount
- D. Cannot determine

Step 1 Chef A: $2 \div 8 = 0.25$ cups per cracker.

Step 2 Chef B: $3 \div 9 \approx 0.33$ cups per cracker.

Step 3 Chef B uses more cheese per cracker.



Answer: A. Chef B

 We do — together

Solve this one with your class using the same steps.

Problem: Which ratio is greater: 3:4 or 5:8?

- A. 3:4
- B. 5:8
- C. They are equal
- D. Cannot determine

Step 1

Step 2

Step 3

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Which lemonade recipe is more lemony? Recipe A: 2 lemons for 6 cups of water. Recipe B: 3 lemons for 7 cups of water.

- A. Recipe B
- B. Recipe A
- C. They are the same
- D. Cannot determine

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Chef Reyes uses 3 tbsp cocoa for every 5 oz of milk, and Chef Tran uses 4 tbsp for every 7 oz. Can you tell whose cocoa is more chocolatey just by looking at the numbers? Why or why not?

Level 1 support

Start here: You can't compare fairly until the ratios share the same amount of milk or you find cocoa per 1 oz.

Need a hint?

Sentence starters (optional)

You ____ tell just by looking because the ____ amounts are different.

____ se puede saber solo mirando porque las cantidades de ____ son diferentes.

To compare fairly we need the same ____ or the cocoa per ____.

Para comparar de forma justa necesitamos la misma ____ o el cacao por ____.

Word bank: ratio compare unit rate equivalent ratio fair

Listen for: Students recognize that 3:5 and 4:7 have different milk amounts, so a fair comparison needs equal milk (a common amount) or a unit rate (cocoa per 1 oz).

Level 2

Some students guess Chef Tran is more chocolatey because 4 is more than 3. Why is that reasoning not enough? Justify using the milk amounts.

- Looking only at the cocoa is not enough because ____.
- More cocoa does not always mean more chocolatey since ____.

EXPLORE

You scaled both recipes to 35 oz of milk: Chef Reyes became 21:35 and Chef Tran became 20:35. How does scaling to a common denominator let you compare the recipes fairly?

Level 1 support

Start here: When both ratios share 35 oz of milk, you only have to compare the cocoa amounts.

 **Need a hint?**

Sentence starters (optional)

After scaling to 35 oz, Chef Reyes has ____ tbsp and Chef Tran has ____ tbsp.

Después de escalar a 35 oz, Chef Reyes tiene ____ cucharadas y Chef Tran tiene ____ cucharadas.

Now I can compare because the ____ is the same, so I just compare the ____.

Ahora puedo comparar porque la ____ es igual, así que solo comparo el ____.

Word bank: equivalent ratio common denominator compare simplify milk

Listen for: Students explain that equal milk (35 oz) makes the comparison fair, so $21 \text{ tbsp} > 20 \text{ tbsp}$ means Chef Reyes is more chocolatey.

Level 2

Find the unit rate (cocoa per 1 oz) for each chef and show it gives the same winner as the 35 oz scaling. Why must both methods agree?

- The unit rates are about ____ and ____ tbsp per oz, so ____ wins.
- Both methods agree because ____.

CONNECT

Two pizza shops have deals: Mario's sells 2 large pizzas for \$18, and Sal's sells 3 for \$25. How would you compare these to find the best price per pizza?

Level 1 support

Start here: Find the unit rate (price per 1 pizza) for each shop, then compare those rates.



Need a hint?

Sentence starters (optional)

Mario's costs ____ per pizza because 18 divided by 2 equals ____.

Mario's cuesta ____ por pizza porque 18 entre 2 es ____.

The better deal is ____ because its ____ is lower.

La mejor oferta es ____ porque su ____ es más baja.

Word bank: unit rate compare ratio better deal per

Listen for: Students compute \$9.00/pizza for Mario's and about \$8.33/pizza for Sal's, then choose Sal's because the lower unit rate is the better deal.

Level 2

If you only need 2 pizzas, does the better unit rate always mean the better total price?

Explain when buying the deal with the lower unit rate might not save money.

- A lower unit rate is the best total deal only when ____.
- It might not save money if ____.

Write About the Math

The Writing Revolution

I can explain my comparison using the words unit rate, equivalent ratio, compare, and common denominator.

1. Kernel Sentence subject + verb

Model: Compare is to look at ratios and see which is bigger, smaller, or equal.

Comparar es mirar razones para ver cuál es mayor, menor o igual.

Write a kernel sentence about compare. Use a subject and a verb.

Escribe una oración base sobre comparar. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Compare matters in math

Comparar importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Compare matters in math because ____.

Comparar importa en matemáticas porque ____.

but
pero

Compare matters in math, but ____.

Comparar importa en matemáticas, pero ____.

so
entonces

Compare matters in math, so ____.

Comparar importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about compare.
Di un hecho verdadero sobre compare.

Compare ____.

Question *Pregunta*

Ask a question about compare.
Haz una pregunta sobre compare.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about compare.
Muestra entusiasmo sobre compare.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with compare.
Dile a un compañero qué hacer con compare.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

You ____ tell just by looking because the ____ amounts are different.

____ se puede saber solo mirando porque las cantidades de ____ son diferentes.

To compare fairly we need the same ____ or the cocoa per ____.

Para comparar de forma justa necesitamos la misma ____ o el cacao por ____.

After scaling to 35 oz, Chef Reyes has ____ tbsp and Chef Tran has ____ tbsp.

Después de escalar a 35 oz, Chef Reyes tiene ____ cucharadas y Chef Tran tiene ____ cucharadas.

Try It

Solve on your own. Check the answer key when you are done.

1. Recipe X uses 2 cups sugar per 8 cookies. Recipe Y uses 3 cups per 10 cookies. Which is sweeter per cookie?

- A. Recipe Y
- B. Recipe X
- C. Same sweetness
- D. Cannot tell

Show your work:

2. Team A won 6 of 10 games. Team B won 9 of 15 games. Which team has the better win ratio?

- A. They are equal
- B. Team A
- C. Team B
- D. Cannot tell

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Store A sells 5 notebooks for \$12. Store B sells 8 notebooks for \$20. Without calculating, a student says Store B must be cheaper because you get more notebooks. Is the student's reasoning valid? Find the unit rates to prove your answer.

Show your work:

Reflect — Exit Ticket

A bakery sells 4 muffins for \$10 and another bakery sells 6 muffins for \$14. Which bakery has the lower price per muffin?

- A. The second bakery
- B. The first bakery
- C. Same price
- D. Cannot determine

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $3:4$ — *Convert to the same denominator: $3:4 = 6:8$. Since $6:8 > 5:8$, the ratio $3:4$ is greater.*
2. **You Try (notes):** A. Recipe B — *Recipe A: $2 \div 6 \approx 0.33$ lemons per cup. Recipe B: $3 \div 7 \approx 0.43$ lemons per cup. Since $0.43 > 0.33$, Recipe B is more lemony.*
3. **Try It 1:** A. Recipe Y — *$X = 0.25$ cup/cookie; $Y = 0.30$ cup/cookie. Y has more sugar per cookie.*
4. **Try It 2:** A. They are equal — *$6:10 = 3:5$ and $9:15 = 3:5$, so the win ratios are equal.*
5. **Exit Ticket:** A. The second bakery — *First bakery: $\$10 \div 4 = \2.50 per muffin. Second bakery: $\$14 \div 6 \approx \2.33 per muffin. The second bakery has the lower price.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Compare is to look at ratios and see which is bigger, smaller, or equal.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).